

2015
YEAR 12
TRIAL EXAMINATION

Mathematics Extension 2

General Instructions

- Reading time - 5 minutes
- Working time - 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- Show all necessary working in Questions 11-16

Total marks - 100

Section I

10 marks

- Attempt Questions 1-10
- Allow about 15 minutes for this section

Section II

90 marks

- Attempt Questions 11-16
- Allow about 2 hour 45 minutes for this section

Section 1

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10.

1 $\frac{2i}{1-2i}=?$

(A) $\frac{4}{3}-\frac{2}{3}i$

(B) $-\frac{4}{5}+\frac{2}{5}i$

(C) $\frac{4}{5}+\frac{2}{5}i$

(D) $-\frac{4}{3}-\frac{2}{3}i$

2 The eccentricity of the hyperbola $\frac{x^2}{4k^2}-\frac{y^2}{k^2}=1$, where k is a positive constant, is?

(A) $\frac{\sqrt{3}}{2}$

(B) 2

(C) $\frac{\sqrt{5}}{2}$

(D) $\sqrt{5}$

- 3 The value of $\int_{-1}^2 \frac{1}{x^2 + 2x + 10} dx$ is?
- (A) $\frac{\pi}{12}$
- (B) $\frac{\pi}{4}$
- (C) $\frac{\pi}{36}$
- (D) None of the above.
- 4 The gradient of the curve $xy - x^2 + 3 = 0$ at the point when $x = 1$ is:
- (A) -4 (B) -1 (C) 1 (D) 4
- 5 The region bounded by the curves $y = x^2$ and $y = x^3$ in the first quadrant is rotated about the y -axis. The volume of the solid of revolution formed can be found using:
- (A) $V = \pi \int_0^1 \left(y^{\frac{1}{3}} - y^{\frac{1}{2}} \right) dy$ (B) $V = \pi \int_0^1 \left(y^{\frac{1}{2}} - y^{\frac{1}{3}} \right) dy$
- (C) $V = \pi \int_0^1 \left(y^{\frac{2}{3}} - y \right) dy$ (D) $V = \pi \int_0^1 (x^4 - x^6) dx$
- 6 What is the remainder when $P(x) = x^3 + x^2 - x + 1$ is divided by $(x - 1 - i)$?
- (A) $-3i - 2$ (B) $3i - 2$ (C) $3i + 2$ (D) $2 - 3i$
- 7 The value of $\lim_{n \rightarrow \infty} \left[n \sin \left(\frac{2\pi}{n} \right) \right]$ is?
- (A) $\frac{1}{2\pi}$
- (B) 1
- (C) 0
- (D) 2π

8 Solve the inequality: $\frac{x+1}{x-3} \leq \frac{x+3}{x-2}$

(A) $x < 2$ and $x > 3$

(B) $x < 2$ and $3 < x \leq 7$

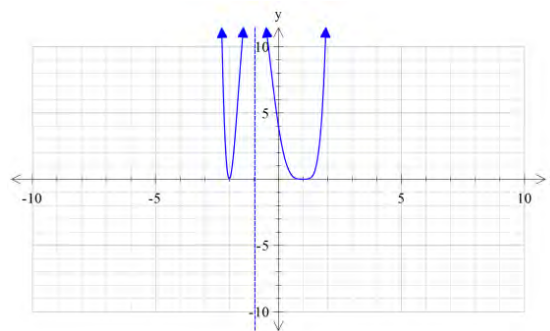
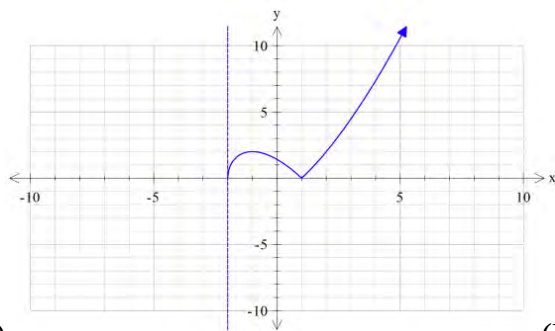
(C) $2 < x < 3$

(D) $2 < x < 3$ and $x \geq 7$

9 Which of the diagrams below best represents $y = \sqrt{f(x)}$

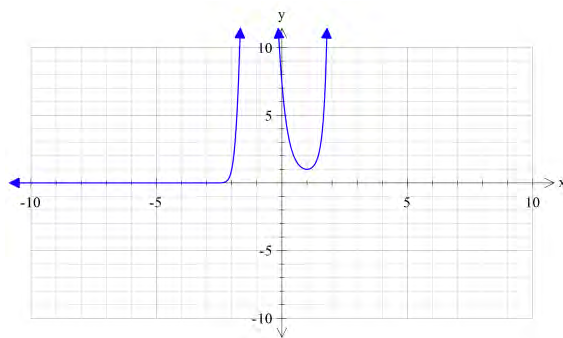
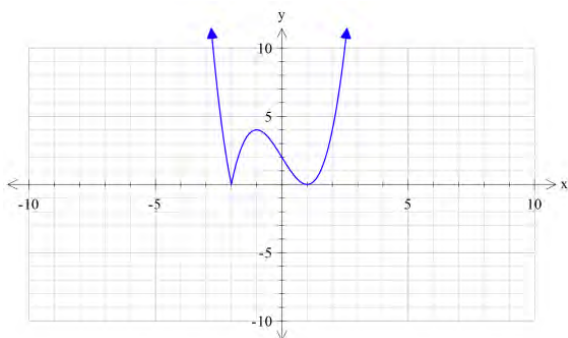
(A)

(B)

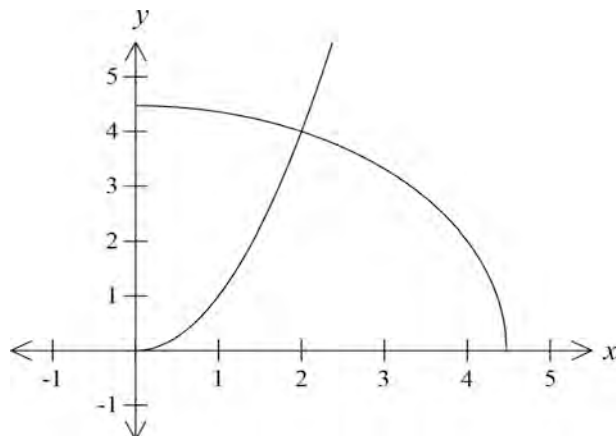


(C)

(D)



10. A solid is formed when the region bounded by the curves $y = x^2$, $y = \sqrt{20 - x^2}$ and the y -axis is rotated about the y -axis.



What is the correct expression for the volume of this solid using the method of cylindrical shells?

- (A) $V = \int_0^2 2\pi \left(\sqrt{20 - x^2} - x^2 \right) dx$
- (B) $V = \int_0^2 2\pi x \left(\sqrt{20 - x^2} - x^2 \right) dx$
- (C) $V = \int_0^2 2\pi \left(x^2 - \sqrt{20 - x^2} \right) dx$
- (D) $V = \int_0^2 2\pi x \left(x^2 - \sqrt{20 - x^2} \right) dx$

End of Section I

Section II

90 marks

Attempt Questions 11 – 16

Allow about 2 hours and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11 – 16 your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

(a) (i) Write $(1 + 2i)^2$ in the form $x + iy$ where x and y are real. 1

(ii) Solve $z^2 = -12 + 16i$. Write your answer in the form $x + iy$. 2

(b) Evaluate $\int_0^1 \frac{2x+1}{x^2+1} dx$. 3

(c) (i) Find A and B such that $\frac{4}{4-x^2} \equiv \frac{A}{2-x} + \frac{B}{2+x}$. 2

(ii) Hence find $\int \frac{4}{4-x^2} dx$ 2

(d) The equation $2x^3 + 4x - 3 = 0$ has roots $x = \alpha$, $x = \beta$ and $x = \gamma$. 2

Find a polynomial equation with roots $x = 2\alpha$, $x = 2\beta$ and $x = 2\gamma$.

(e) Evaluate $\int_0^{\frac{\pi}{2}} \cos \theta \sqrt{1 + \sin \theta} d\theta$. 3

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet.

- (a) The equation $x^4 + 2x^3 - 7x^2 - 20x - 12 = 0$ has a double root. Find this root and hence solve this equation. 3
- (b) Prove that the condition for the line $y = mx + c$ to touch the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is:
$$c^2 = b^2 + a^2 m^2$$
 4
- (c) Given that $1 + i$ is a root of the equation $z^2 + (a + 2i)z + (5 + ib) = 0$ where **a** and **b** are real, determine the values of **a** and **b**. 3
- (d) (i) Find the equation of the tangent at the point $P\left(ct, \frac{c}{t}\right)$ on the rectangular hyperbola $xy = c^2$. 2
- (ii) Find the coordinates of A and B where this tangent cuts the x and y axis respectively. 2
- (iii) Prove that the area of the triangle OAB is a constant. (Where O is the origin). 1

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

(a) Let $I_n = \int_0^{\frac{\pi}{4}} \tan^n x \, dx$, where n is an integer and $n \geq 0$.

(i) Show that $I_n + I_{n-2} = \frac{1}{n-1}$. 3

(ii) Hence find $\int_0^{\frac{\pi}{4}} \tan^4 x \, dx$. 2

(b) $P(a \cos \theta, b \sin \theta)$, where $0 < \theta < \frac{\pi}{2}$, is a point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

(i) Show that the tangent at P has equation 2

$$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1.$$

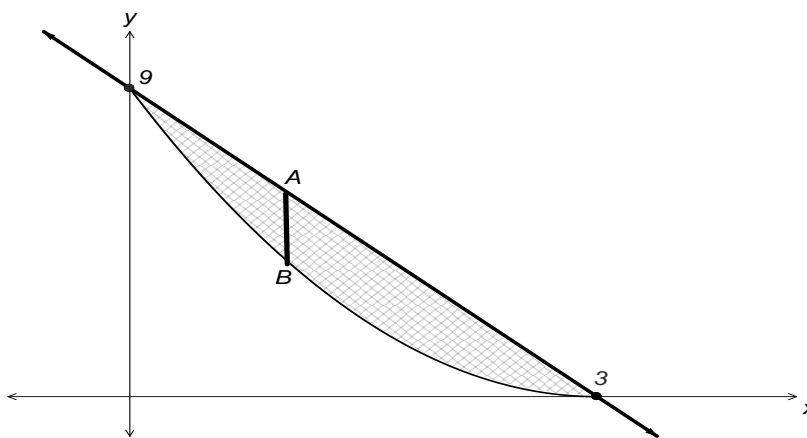
(ii) If L is the distance of the tangent from the origin O 3

show that $L > \frac{ab}{\sqrt{a^2 + b^2}}$.

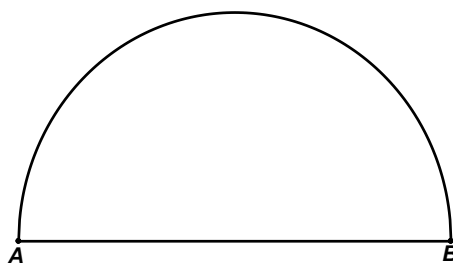
Question 13 continues over the page

Question 13 continued

(c)



Cross-section with base on AB



The diagram above shows the region enclosed by the parabola $y = (x-3)^2$ and the line $3x + y = 9$. The region forms the base of a solid.

When the solid is sliced perpendicular to the x -axis, each cross-section is a semi-circle with diameter across the region.

A typical cross-section is shown above.

(i) If the solid is sliced along the line $x = a$, show that the area of 2

the cross-section is $A = \frac{\pi}{8} a^2 (3-a)^2$, where $0 \leq a \leq 3$.

(ii) Find the volume of the solid. 3

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

- (a) Using the substitution $t = \tan \frac{x}{2}$, or otherwise, evaluate 3

$$\int_0^{\frac{\pi}{2}} \frac{1}{1 + \cos x} dx$$

- (b) A sequence is defined such that $u_1 = 1, u_2 = 1$ and $u_n = u_{n-1} + u_{n-2}$ for $n \geq 3$. 4

Prove by induction that $u_n < \left(\frac{7}{4}\right)^n$ for integers $n \geq 1$.

- (c) A variable point $P(x, y)$ moves so that its distance from $(0, 1)$ is one-half its distance from $y = 4$. Find, and describe the locus of P . 3

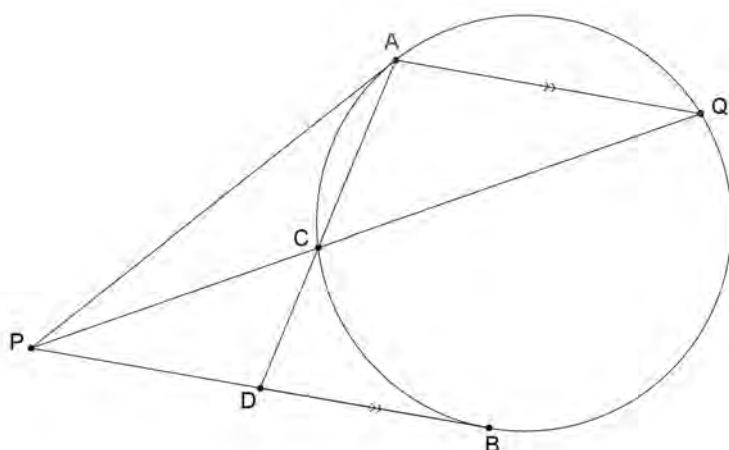
- (d) If $a > 0, b > 0, c > 0$ and $a + b + c = 1$ show that $(1-a)(1-b)(1-c) \geq 8abc$. 2

- (e) Show that $\int e^{-x} \cos x dx = \frac{1}{2}(\sin x - \cos x) + C$ 3

End of Question 14

Question 15 (15 marks) Use a SEPARATE writing booklet.

- (a) In the diagram below, PA and PB are tangents to the circle. The chord AQ is parallel to the tangent PB . PCQ is a secant to the circle and chord AC produced meets PB at D .
- (i) Show that $\triangle CDP$ is similar to $\triangle PDA$. 2
- (ii) Hence show that $PD^2 = AD \times CD$. 1
- (iii) Hence, or otherwise, prove that AD bisects PB . 2



- (b) (i) Three identical balls are to be placed randomly in three trays. 2
Each ball is equally likely to be placed in any one of the trays.
Show that the probability that exactly one of the trays is empty
is $\frac{2}{3}$.
- (ii) The above process is repeated with n identical balls (where $n \geq 2$) 2
and n trays. Write an expression in terms of n for the probability that
exactly one tray is empty.

Question 15 continues over the page

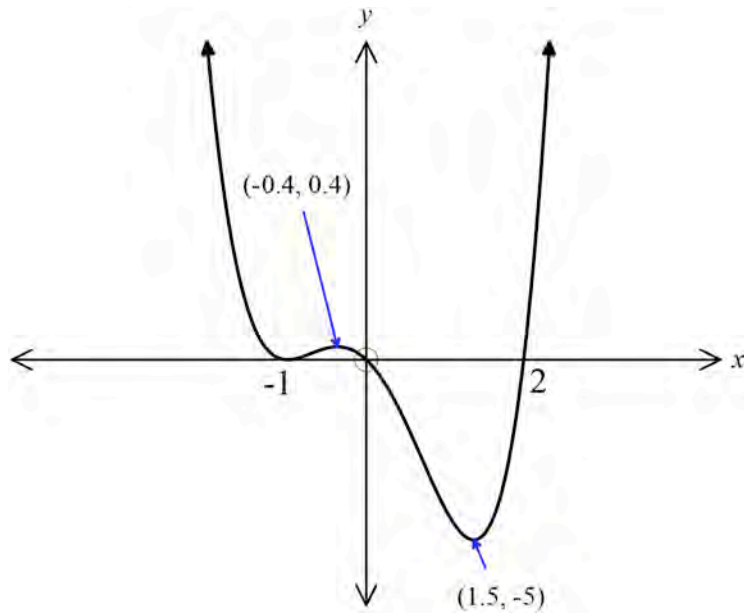
Question 15 continued

- (c) Use the method of cylindrical shells to find the volume of the solid generated by revolving the region enclosed by $y = 3x^2 - x^3$ and the x axis around the y -axis. **3**
- (d) By taking slices perpendicular to the axis of rotation, find the volume when The area bounded by the parabola $y = 6x - x^2$ and the x axis is rotated about the line $y = 10$ **3**

End of Question 15

Question 16 (15 marks) Use a SEPARATE writing booklet.

- (a) The graph of $y = f(x)$ is shown below.



Draw separate sketches for each of the following:

- | | | |
|-------|----------------------|---|
| (i) | $y = f(x) $ | 2 |
| (ii) | $y = \frac{1}{f(x)}$ | 2 |
| (iii) | $y^2 = f(x)$ | 2 |
| (iv) | $y = e^{f(x)}$ | 2 |

(b) The function $F(p)$ is defined as $F(p) = \lim_{t \rightarrow \infty} \int_0^t x^{p-1} e^{-x} dx$, for $p > 0$.

(i) Show that $F(1) = 1$. 2

(ii) Use integration by parts to show $F(p+1) = pF(p)$. 3

(iii) Hence find $F(n)$ for integers $n \geq 1$. 2

End of paper.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

Question MULTIPLE CHOICE

$$\begin{aligned} \textcircled{1} \quad \frac{2i}{1-2i} \times \frac{1+2i}{1+2i} &= \frac{2i-4}{1+4} \\ &= -\frac{4}{5} + \frac{2i}{5} \end{aligned}$$

B

$$\begin{aligned} \textcircled{2} \quad \frac{x^2}{4k^2} - \frac{y^2}{k^2} &= 1 & k^2 &= 4k^2(e^2-1) \\ \frac{1}{4} &= e^2-1 \\ e^2 &= \frac{5}{4} \\ e &= \frac{\sqrt{5}}{2} \end{aligned}$$

C

$$\begin{aligned} \textcircled{3} \quad \int_{-1}^2 \frac{1}{x^2+2x+10} dx &= \int_{-1}^2 \frac{1}{x^2+2x+1+9} dx \\ &= \int_{-1}^2 \frac{1}{(x+1)^2+9} dx = \frac{1}{3} \tan^{-1} \frac{x+1}{3} \\ &= \frac{1}{3} [\tan^{-1} 1 - 0] \\ &= \frac{1}{3} \times \frac{\pi}{4} = \frac{\pi}{12} \end{aligned}$$

A

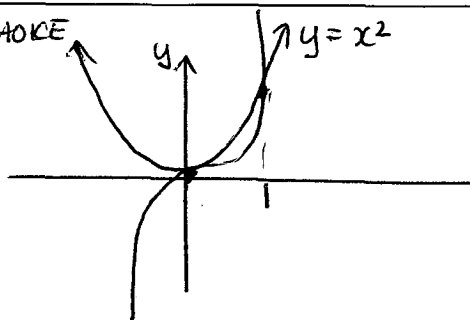
$$\begin{aligned} \textcircled{4} \quad 1 \cdot y + 1 \cdot \frac{dy}{dx} \cdot x - 2x &= 0 \\ \text{at } (1, -2) & \\ -2 + \frac{dy}{dx} - 2 &= 0 \\ \frac{dy}{dx} &= 4 \end{aligned}$$

$$\begin{aligned} x &= 1 \\ y &= -2 \end{aligned}$$

D

Question MULTICHOICE

⑤



$$\begin{aligned}
 V &= \pi \int_0^1 x^2 dy \\
 V &= \pi \int_0^1 (y^{3/2})^2 - (y^{1/2})^2 dy \\
 &= \pi \int_0^1 y^{3/2} - y^{1/2} dy
 \end{aligned}$$

C

⑥ $P(x) = x^3 + x^2 - x + 1$ $(x - 1 - i)$
 $x - (1 + i)$

$$\begin{aligned}
 P(1+i) &= (1+i)^3 + (1+i)^2 - (1+i) + 1 \\
 &= 1 + 3i - 3 - i + 1 + 2i - 1 - 1 + i + 1 \\
 &= -2 + 3i
 \end{aligned}$$

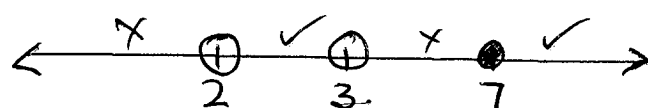
B

⑦ $\lim_{n \rightarrow \infty} \left[n \sin \left(\frac{2\pi}{n} \right) \right]$ * Small angle.
 $\sin \left(\frac{2\pi}{n} \right) = \frac{2\pi}{n}$
 $= \left[\cancel{n} \times \frac{2\pi}{\cancel{n}} \right]$

D

⑧ Solve $(x+1)(x-2) = (x+3)(x-3)$
 $x^2 - x - 2 = x^2 - 9$

$$\begin{aligned}
 -x &= -7 \\
 x &= 7
 \end{aligned}$$



for $x=0$ ($\Rightarrow x=2$) $\frac{1}{-3} \leq \frac{3}{-2}$ $\frac{1}{3} \geq \frac{3}{2} \times$

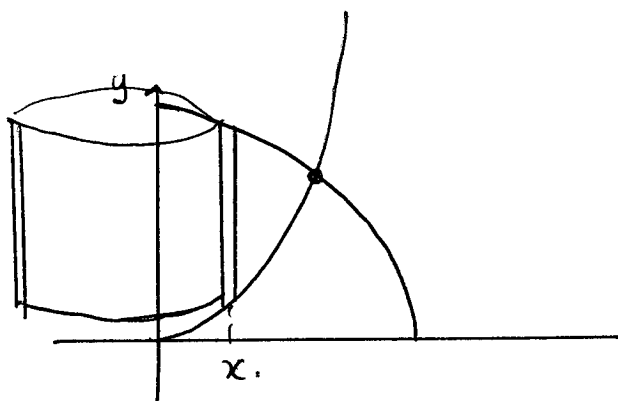
$$2 < x < 3, x \geq 7$$

D

Question MULTIPLE CHOICE

9 $y = \sqrt{f(x)}$

10



$$2\pi r = 2\pi x$$

$$A = 2\pi x (\sqrt{20-x^2} - x^2)$$

$$\begin{matrix} y_1 - y_2 \\ \sqrt{20-x} - x^2 \end{matrix}$$

$$V = \int_0^2 2\pi x (\sqrt{20-x^2} - x^2) dx$$

A

B

Question 11

$$(a) (i) (1+2i)^2 = 1+4i-4$$

$$= \boxed{-3+4i}$$

1

$$(ii) z^2 = -12+16i$$

$$(x+iy)^2 = -12+16i$$

$$x^2+2xyi-y^2 = -12+16i$$

$$x^2-y^2 = -12 \quad \dots \textcircled{1}$$

$$2xy = 16$$

$$xy = 8 \rightarrow y = \frac{8}{x}$$

$\therefore$ sub into $\textcircled{1}$

$$x^2 - \frac{64}{x^2} = -12$$

$$x^4 - 64 = -12x$$

$$x^4 + 12x - 64 = 0$$

$$(x^2+16)(x^2-4) = 0 \quad x \text{ is real.}$$

$$\therefore x = \pm 2$$

$$y = \pm 4$$

$$\therefore \boxed{z = 2-4i, -2-4i}$$

2

Question 11

$$\begin{aligned}
 \text{(b)} \quad \int_0^1 \frac{2x+1}{x^2+1} &= \int_0^1 \frac{2x}{x^2+1} + \frac{1}{x^2+1} dx \\
 &= \ln(x^2+1) \Big|_0^1 + \tan^{-1} x \Big|_0^1 \\
 &= \ln(2) - \ln(1) + \tan^{-1} 1 - 0 \\
 &= \boxed{\ln 2 + \frac{\pi}{4}}
 \end{aligned}$$

3

$$\begin{aligned}
 \text{(c) (i)} \quad \frac{4}{4-x^2} &= \frac{A}{2-x} + \frac{B}{2+x} \\
 4 &= A(2+x) + B(2-x) \\
 &\quad \text{for } x = -2 \\
 4 &= B(4) \rightarrow \boxed{B=1} \\
 &\quad \text{for } x = 2 \\
 4 &= A(4) \rightarrow \boxed{A=1}
 \end{aligned}$$

2

$$\begin{aligned}
 \text{(ii)} \quad \int \frac{4}{4-x^2} dx &= \int \frac{1}{2-x} + \frac{1}{2+x} dx \\
 &= -\ln(2-x) + \ln(2+x) + c \\
 &= \boxed{\ln \left| \frac{2+x}{2-x} \right| + c}
 \end{aligned}$$

2

Question 11

$$(d) \quad 2x^3 + 4x - 3 = 0. \quad \alpha, \beta, \gamma$$

$$x = 2\alpha$$

$$\alpha = \frac{x}{2}$$

$$\therefore \frac{2x^3}{8} + \frac{4x}{2} - 3 = 0$$

$$x^3 + 8x - 12 = 0$$

2

$$(e) \quad \int_0^{\frac{\pi}{2}} \cos \theta \sqrt{1 + \sin \theta} \, d\theta$$

$$\text{let } u = 1 + \sin \theta$$

$$du = \cos \theta \, d\theta$$

$$= \int_1^2 u^{\frac{1}{2}} \, du$$

$$= \left[\frac{2}{3} u^{\frac{3}{2}} \right]_1^2$$

$$= \frac{2}{3} [2\sqrt{2} - 1]$$

$$= \frac{4\sqrt{2} - 3}{3}$$

3

Question 12

$$(a) P(x) = x^4 + 2x^3 - 7x^2 - 20x - 12$$

$$P'(x) = 4x^3 + 6x^2 - 14x - 20.$$

$$P'(-2) = 0 \quad P(-2) = 0$$

$\therefore$ Double root is 2.

$$(x+2)^2 = x^2 + 4x + 4$$

$$\begin{array}{r} x^2 - 2x - 3 \\ x^2 + 4x + 4 \overline{) x^4 + 2x^3 - 7x^2 - 20x - 12} \\ \underline{x^4 + 4x^3 + 4x^2} \\ -2x^3 - 11x^2 - 20x - 12 \\ \underline{-2x^3 - 8x^2 - 8x} \\ -3x^2 - 12x - 12 \\ \underline{-3x^2 - 12x - 12} \\ 0 \end{array}$$

$$-2x^3 - 11x^2 - 20x - 12$$

$$\underline{-2x^3 - 8x^2 - 8x}$$

$$-3x^2 - 12x - 12$$

$$\underline{-3x^2 - 12x - 12}$$

$$P(x) = (x+2)^2(x-3)(x+1)$$

$$\boxed{x = -2, -2, 3, -1}$$

$$(b) y = mx + c \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\therefore \frac{x^2}{a^2} + \frac{(mx+c)^2}{b^2} = 1$$

$$b^2x^2 + a^2m^2x^2 + 2a^2mcx + a^2c^2 = a^2b^2$$

$$(b^2 + a^2m^2)x^2 + 2a^2mcx + a^2c^2 - a^2b^2 = 0$$

$$\text{For } \Delta = 0$$

① Finding double root $x = 2$

① Long division or equivalent

① Finding other roots.

3

① attempting to solve simultaneously

① $\Delta = 0$ or equivalent

① Algebraic manipulation with minor error.

Question 12

$$4a^4m^2c^2 - 4(b^2 + a^2m^2)(a^2c^2 - a^2b^2) = 0$$

$$4a^4m^2c^2 - 4a^2b^2c^2 + 4a^2b^4 - 4a^4m^2c^2 + 4a^4m^2b^2 = 0$$

$$-4a^2b^2c^2 + 4a^2b^4 + 4a^4m^2b^2 = 0$$

$$-c^2 + b^2 + a^2m^2 = 0$$

$$\therefore c^2 = b^2 + a^2m^2$$

as required

① answer.

4

(c) $z^2 + (a+2i)z + (5+ib) = 0$

sub in $z = 1+i$

$$(1+i)^2 + (a+2i)(1+i) + (5+ib) = 0$$

$$1+2i-1 + a+ai+2i-2+5+ib = 0$$

$$a+3 + 2i+ai+2i+bi = 0$$

Equate real & imaginary parts

$$a+3 = 0$$

$$\therefore a = -3$$

$$4+a+b = 0$$

$$4-3+b = 0$$

$$1+b = 0$$

$$b = -1$$

$$\therefore \boxed{a=-3, b=-1}$$

① subbing
into equation① correct
expansion① Equating
real /
imaginary
parts**3**

Question 12

$$\begin{aligned} \text{(d) (i)} \quad xy &= c^2 \\ y &= c^2 x^{-1} \\ y' &= -c^2 x^{-2} \\ \therefore m_T &= \frac{-c^2}{c^2 t^2} \end{aligned}$$

$$= -\frac{1}{t^2}$$

$$\therefore y - \frac{c}{t} = -\frac{1}{t^2} (x - ct)$$

$$t^2 y - ct = -x + ct$$

$$\boxed{x + t^2 y = 2ct}$$

$$\text{(ii) at A } y = 0$$

$$x = 2ct \quad \therefore \boxed{A = (2ct, 0)}$$

$$\text{at B } x = 0$$

$$t^2 y = 2ct$$

$$y = \frac{2c}{t}$$

$$\therefore \boxed{B = (0, \frac{2c}{t})}$$

$$\text{(iii)}$$

$$A = \frac{1}{2} \times 2ct \times \frac{2c}{t}$$

$$\boxed{A = 2c^2}$$

which is a constant.

① correct gradient with correct working

① correct equation

2

① $A = (2ct, 0)$

① $B = (0, \frac{2c}{t})$

2

1

Question 13

$$(a) \quad I_n = \int_0^{\frac{\pi}{4}} \tan^n x \, dx$$

$$(i) \quad I_n + I_{n-2} = \frac{1}{n-1}$$

$$\int_0^{\frac{\pi}{4}} \tan^n x \, dx = \int_0^{\frac{\pi}{4}} \tan^2 x \tan^{n-2} x \, dx$$

$$= \int_0^{\frac{\pi}{4}} (\sec^2 x - 1) \tan^{n-2} x \, dx$$

$$I_n = \int_0^{\frac{\pi}{4}} \sec^2 x \tan^{n-2} x \, dx - \int_0^{\frac{\pi}{4}} \tan^{n-2} x \, dx$$

$$I_n = \int_0^1 u^{n-2} \, du - I_{n-2}$$

$$u = \tan x \\ du = \sec^2 x \, dx$$

$$I_n = \left[\frac{u^{n-1}}{n-1} \right]_0^1 - I_{n-2}$$

$$I_n = \frac{1}{n-1} - 0 - I_{n-2}$$

$$\therefore I_n + I_{n-2} = \frac{1}{n-1}$$

as required.

① correct
set leading
to correct
answer

① correct
integration
(using parts)

① correct
manipulation
leading to
answer

3

Suggested Solutions

Comments

Question 13

$$(ii) \int_0^{\frac{\pi}{4}} \tan^4 x \, dx$$

$$I_4 = \frac{1}{3} - I_2$$

$$I_4 = \frac{1}{3} - \left[\frac{1}{1} - I_0 \right]$$

$$= \frac{1}{3} - \left[1 - \frac{\pi}{4} \right]$$

$$\boxed{I_n = \frac{\pi}{4} - \frac{2}{3}}$$

$$\int_0^{\frac{\pi}{4}} \tan^0 x \, dx,$$

$$= \left[\frac{\pi}{4} - 0 \right]$$

① Correct sub into formula

① Correct use of formula or finding I_2 and or I_0 or correct answer $\boxed{2}$

$$(b) P(a \cos \theta, b \sin \theta) \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$(i) \frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0$$

$$\frac{2y}{b^2} \frac{dy}{dx} = -\frac{2x}{a^2}$$

$$\frac{dy}{dx} = -\frac{x}{a^2} \times \frac{b^2}{y}$$

$$m_1 = -\frac{a \cos \theta \cdot b^2}{a^2 b \sin \theta}$$

$$= -\frac{b \cos \theta}{a \sin \theta}$$

① Correct gradient with working

Question 13

$$\therefore y - b \sin \theta = -\frac{b \cos \theta}{a \sin \theta} (x - a \cos \theta)$$

$$a \sin \theta y - ab \sin^2 \theta = -b \cos \theta x + ab \cos^2 \theta$$

$$a \sin \theta y + b \cos \theta x = ab$$

$$\frac{\sin \theta y}{b} + \frac{\cos \theta x}{a} = 1$$

as required.

① correct manipulation to get required equation

2

$$(ii) d = \frac{|0 + 0 - 1|}{\sqrt{\frac{\cos^2 \theta}{a^2} + \frac{\sin^2 \theta}{b^2}}}$$

$$= \frac{1}{\frac{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}}{ab}}$$

$$= \frac{ab}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}} \quad \text{Since } 0 \leq \sin^2 \theta \leq 1 \text{ and } 0 \leq \cos^2 \theta \leq 1$$

$$b^2 \cos^2 \theta + a^2 \sin^2 \theta < a^2 + b^2$$

$$\therefore \frac{1}{\frac{b^2 \cos^2 \theta + a^2 \sin^2 \theta}{ab}} > \frac{1}{a^2 + b^2}$$

$$\therefore \frac{ab}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}} > \frac{ab}{\sqrt{a^2 + b^2}}$$

as required

① correct expression for $\perp$ distance

① Realising to $0 \leq \sin^2 \theta \leq 1$ and $0 \leq \cos^2 \theta \leq 1$

① Relating the above to come up with required inequality

3

Question 13

(c) (i) $A = \frac{\pi}{2} r^2$

$$= \frac{\pi}{2} \left[\frac{(9-3x) - (x-3)^2}{2} \right]^2$$

$$= \frac{\pi}{2} \left[\frac{\cancel{9} - 3x - x^2 + 6x - \cancel{9}}{2} \right]^2$$

$$= \frac{\pi}{2} \left[\frac{3x - x^2}{2} \right]^2$$

$$= \frac{\pi}{8} [x(3-x)]^2$$

at $x=a$

$$A = \frac{\pi}{8} a^2(3-a)^2$$

as required.

(ii) $V = \int_0^3 \frac{\pi}{8} (9a^2 - 6a^3 + a^4) da$

$$= \frac{\pi}{8} \left[3a^3 - \frac{3a^4}{2} + \frac{a^5}{5} \right]_0^3$$

$$= \frac{\pi}{8} \left[\frac{81}{10} \right]$$

$$\boxed{V = \frac{81\pi}{80} \text{ u}^3}$$

① correct
r

① correct
expression
for A or
correct
expression
with
incorrect
r.

2

① correct
integral
for V

① correct
integration

① correct
sub. and
answer

3

Question 14

$$(a) \int_0^{\frac{\pi}{2}} \frac{1}{1+\cos x} dx$$

$$= \int_0^1 \frac{1}{1+\frac{1-t^2}{1+t^2}} \times \frac{2}{1+t^2} dt$$

$$= \int_0^1 \frac{1+t^2}{1+t^2+1-t^2} \times \frac{2}{1+t^2} dt$$

$$= \int_0^1 1 dt$$

$$= t \Big|_0^1$$

$$= \boxed{1}$$

$$t = \tan \frac{x}{2}$$

$$\frac{x}{2} = \tan^{-1} t$$

$$x = 2 \tan^{-1} t$$

$$dx = \frac{2}{1+t^2} dt$$

- ① correct sub for dx or cos x

- ① correct sub including limits

- ① correct integration and answer

$\boxed{3}$

$$(b) U_1 = 1, U_2 = 1, U_n = U_{n-1} + U_{n-2} \quad n \geq 3$$

For $n=1$

$$U_1 = 1 < \left(\frac{7}{4}\right)^1$$

$\therefore$ true for $n=1$

For $n=2$

$$U_2 = 1 < \left(\frac{7}{4}\right)^2 \Rightarrow 1 < \frac{49}{16}$$

$\therefore$ true for $n=2$

① Prove true for U_1 and U_2

Question 14

Assume true for integers up to $n = k$

$$\therefore U_k < \left(\frac{7}{4}\right)^k$$

Prove true for $n = k+1$

$$\text{R.T.P. } U_{k+1} < \left(\frac{7}{4}\right)^{k+1}$$

$$\begin{aligned} \text{Now } U_{k+1} &= U_k + U_{k-1} \\ &< \left(\frac{7}{4}\right)^k + \left(\frac{7}{4}\right)^{k-1} \\ &= \left(\frac{7}{4}\right)^k + \left(\frac{7}{4}\right)^k \times \frac{4}{7} \\ &= \left(\frac{7}{4}\right)^k \left(1 + \frac{4}{7}\right) \\ &= \left(\frac{7}{4}\right)^k \left(\frac{11}{7}\right) \\ &< \left(\frac{7}{4}\right)^k \left(\frac{7}{4}\right) \\ &= \left(\frac{7}{4}\right)^{k+1} \end{aligned}$$

$$\therefore U_{k+1} < \left(\frac{7}{4}\right)^{k+1}$$

$\therefore$ True by induction

① Assumption
Step - correctly
stated or
RTP step.

① Set up
of proof-
step which
could lead
to correct
solution

① correct
manipulation
and answer

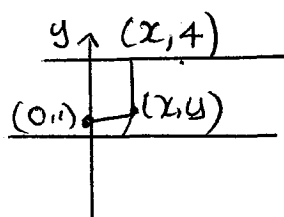
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Suggested Solutions

Comments

Question 14

(c)



$$(x-0)^2 + (y-1)^2 = \frac{1}{4} [(x-x)^2 + (y-4)^2]$$

$$x^2 + y^2 - 2y + 1 = \frac{1}{4} [y^2 - 8y + 16]$$

$$4x^2 + 4y^2 - 8y + 4 = y^2 - 8y + 16$$

$$4x^2 + 3y^2 = 12$$

$$\therefore \boxed{\frac{x^2}{3} + \frac{y^2}{4} = 1}$$

ellipse.

① correct set up or equivalent

① correct algebraic steps

① correct answer including description

3

$$(d) \quad a+b+c=1 \quad (1-a)(1-b)(1-c) \geq 8abc.$$

$$a = 1-b-c$$

$$b = 1-a-c$$

$$c = 1-a-b$$

$$\text{LHS} = (1-1+b+c)(1-1+a+c)(1-1+a+b)$$

$$= (b+c)(a+c)(a+b)$$

$$\geq 2\sqrt{bc} \times 2\sqrt{ac} \times 2\sqrt{ab}$$

$$= 8\sqrt{a^2b^2c^2}$$

$$= 8abc$$

as required.

$$b+c \geq 2\sqrt{bc}$$

$$a+c \geq 2\sqrt{ac}$$

$$a+b \geq 2\sqrt{ab}$$

① Attempt which could lead to a solution

① algebraic manipulation and answer

2

Question 14

$$(e) \quad I_n = \int e^{-x} \cos x \, dx.$$

$$I_n = -e^{-x} \cos x - \int -e^{-x} (-\sin x) \, dx$$

$$= -e^{-x} \cos x - \int e^{-x} \sin x \, dx$$

$$I_n = -e^{-x} \cos x - \left[-e^{-x} \sin x - \int -e^{-x} \cos x \, dx \right]$$

$$I_n = -e^{-x} \cos x + e^{-x} \sin x - I_n + C$$

$$2I_n = -e^{-x} \cos x + e^{-x} \sin x + C$$

$$= e^{-x} (\sin x - \cos x) + C$$

$$\therefore I_n = \frac{1}{2} e^{-x} (\sin x - \cos x) + C$$

as required.

- ① correct use of integration by parts

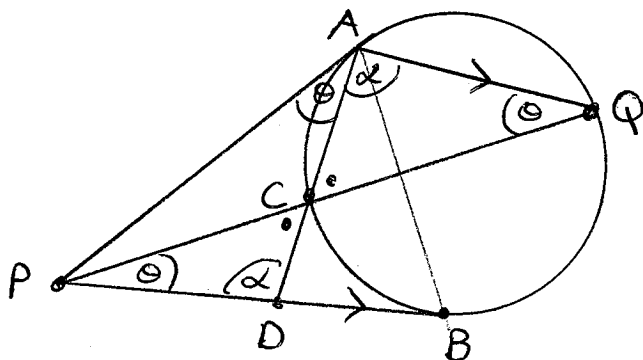
- ① second attempt of integration by parts or equivalent

① Algebraic manipulation and answer.

3

Question 15

(9)

(i) In Δ s CDP and PDA

$$\angle CDP = \angle PDA \text{ (common } \angle)$$

$$\angle PAD = \angle AQP \text{ (alt segment theorem)}$$

$$\angle AQP = \angle CPD \text{ (alt } \angle\text{'s } AQ \parallel PB)$$

$$\therefore \angle PAD = \angle CPD \text{ (both } = \angle AQP)$$

$$\therefore \Delta CDP \parallel \Delta PDA \text{ (AAA)}$$

$$(ii) \quad \frac{PD}{AD} = \frac{CD}{PD}$$

$$\therefore (PD)^2 = AD \times CD$$

as required

$$(iii) \quad (BD)^2 = AD \times CD \text{ (property of tangent and secants)}$$

$$\therefore (BD)^2 = (PD)^2 \rightarrow \text{both} = AD \times CD$$

$$\therefore BD = PD$$

 $\therefore$ AD bisects PB

① Finding 1 equal $\angle$ or equivalent with reasons

① Finding 2nd angle with reasons or equivalent

[2]

[1]

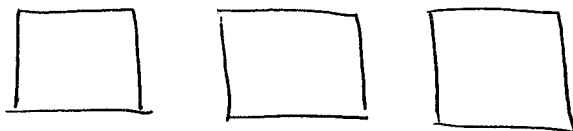
① attempt to use correct theorem

① Equating $BD^2 = PD^2$

[2]

Question 15

(b)



$$\begin{aligned}
 \text{(i) } P(\text{exactly 1 empty}) &= \frac{{}^3C_2 \times 3!}{3^3} \\
 &= \frac{3 \times 6}{27} \\
 &= \frac{2}{3}
 \end{aligned}$$

as required

$$\begin{aligned}
 \text{(ii) } P(\text{exactly 1 empty}) &= \frac{{}^nC_2 \times n!}{n^n} * \\
 &= \frac{n!}{(n-2)! \cdot 2!} \times n! \\
 &\quad \quad \quad n^n \\
 &= \frac{n! \times n!}{2(n-2)! \times n^n} \\
 &= \frac{n! \cdot n(n-1)}{2n^n}
 \end{aligned}$$

① Numerator or denominator correct or minor error in both.

① correct answer from previous step (within reason)

2

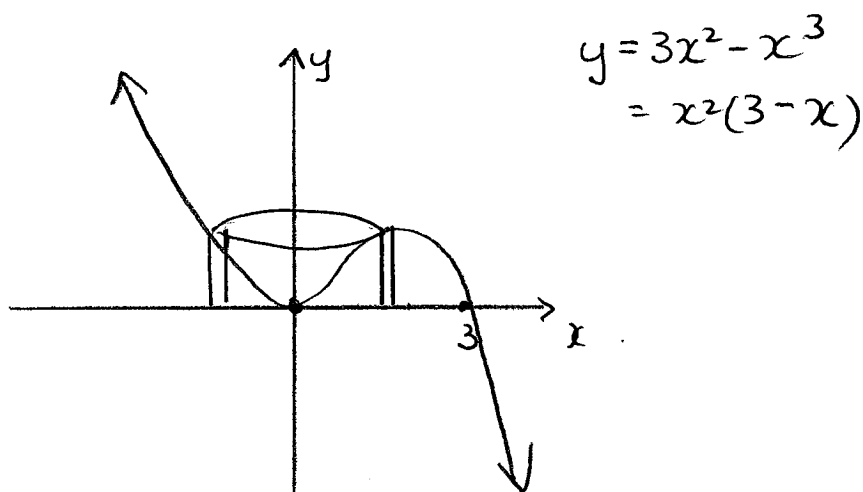
① correct set up or establishing pattern from part (i)

① Any form of correct answer

2

Question 15

(c)



$$2\pi r = 2\pi x$$

$$A = 2\pi x (3x^2 - x^3) \quad y = 3x^2 - x^3$$

$$\Delta V = 2\pi x (3x^2 - x^3) \Delta x$$

$$V = 2\pi \lim_{\Delta x \rightarrow 0} \sum_{i=0}^3 (3x^3 - x^4) \Delta x$$

$$V = 2\pi \int_0^3 (3x^3 - x^4) dx$$

$$= 2\pi \left[\frac{3x^4}{4} - \frac{x^5}{5} \right]_0^3$$

$$= 2\pi \left[\frac{243}{20} \right]$$

$$V = \frac{243\pi}{10} \text{ u}^3$$

① Correct height or radius or equivalent

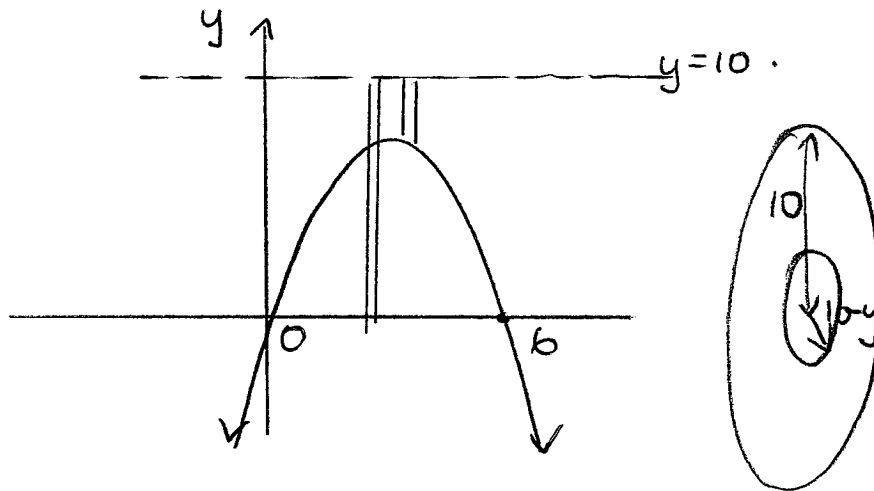
① Correct Set expression for A or correct set up for V from previous steps

① Correct integration and answer

3

Question 15

(d)



$$A = \pi [R^2 - r^2]$$

$$= \pi [10^2 - (10 - y)^2]$$

$$A = \pi [100 - 100 + 20y - y^2]$$

$$= \pi [20y - y^2]$$

$$A(x) = \pi [20(6x - x^2) - (6x - x^2)^2]$$

$$= \pi [120x - 20x^2 - 36x^2 + 12x^3 - x^4]$$

$$= \pi [120x - 56x^2 + 12x^3 - x^4]$$

$$\Delta v = \pi [120x - 56x^2 + 12x^3 - x^4] \Delta x$$

$$V = \pi \lim_{\Delta x \rightarrow 0} \sum_{i=0}^6 (120x - 56x^2 + 12x^3 - x^4) \Delta x$$

$$V = \pi \int_0^6 (120x - 56x^2 + 12x^3 - x^4) dx$$

$$= \pi \left[60x^2 - \frac{56x^3}{3} + 3x^4 - \frac{x^5}{5} \right]_0^6$$

$$= \pi \left[\frac{2304}{5} \right]$$

$$V = \frac{2304\pi}{5} U^2$$

① correct use of washer method

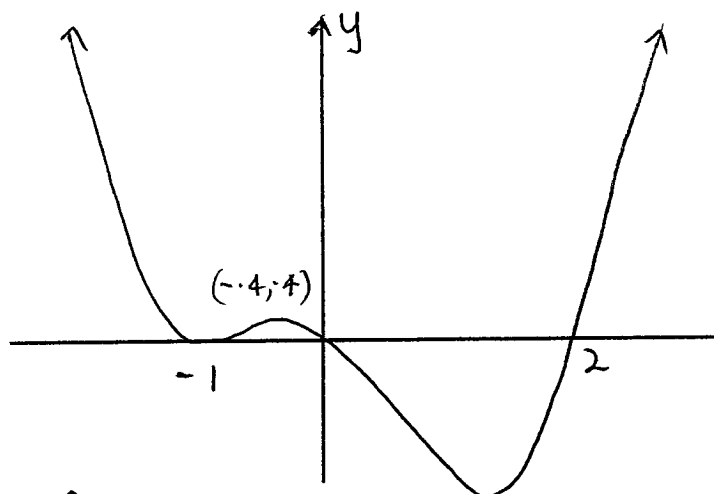
① correct expression for A or Δv or V.

① correct integration of appropriate function and answer

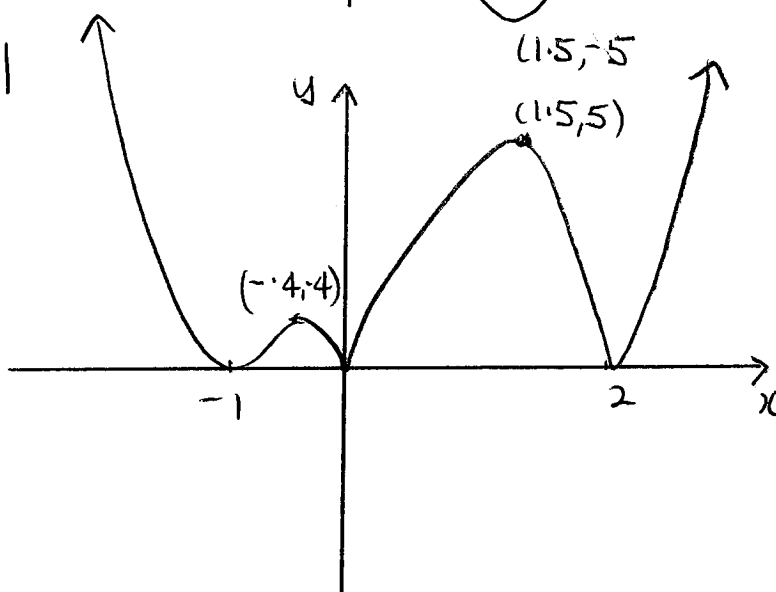
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Question 16

(a)

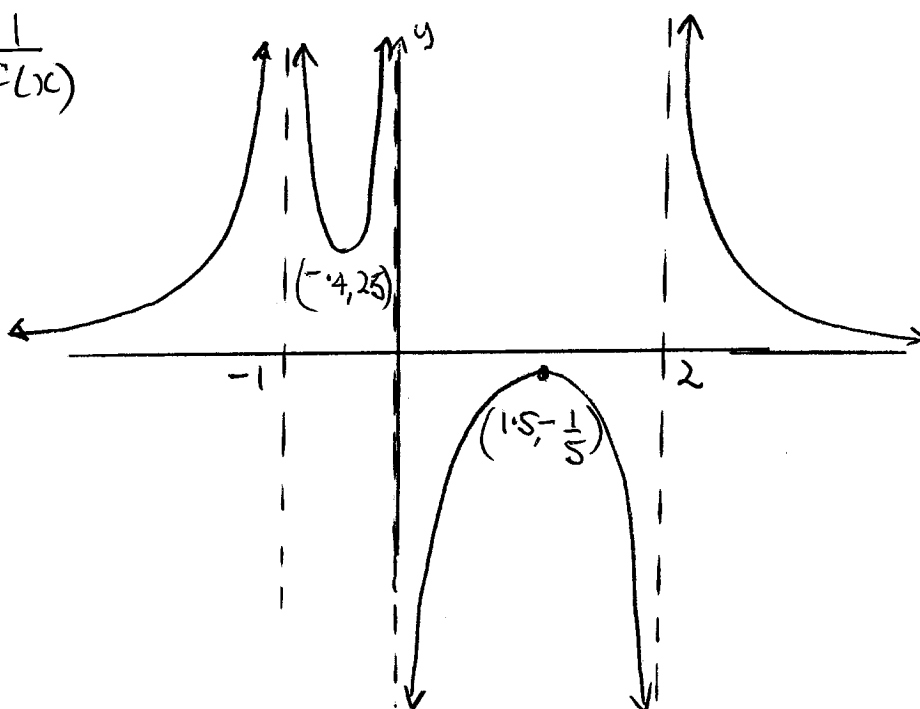


(i) $y = |f(x)|$



2

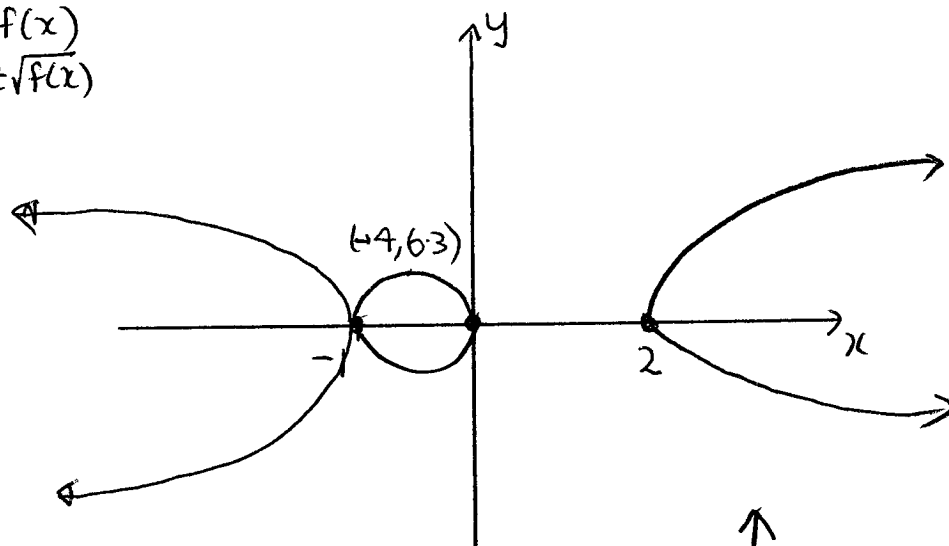
(ii) $y = \frac{1}{f(x)}$



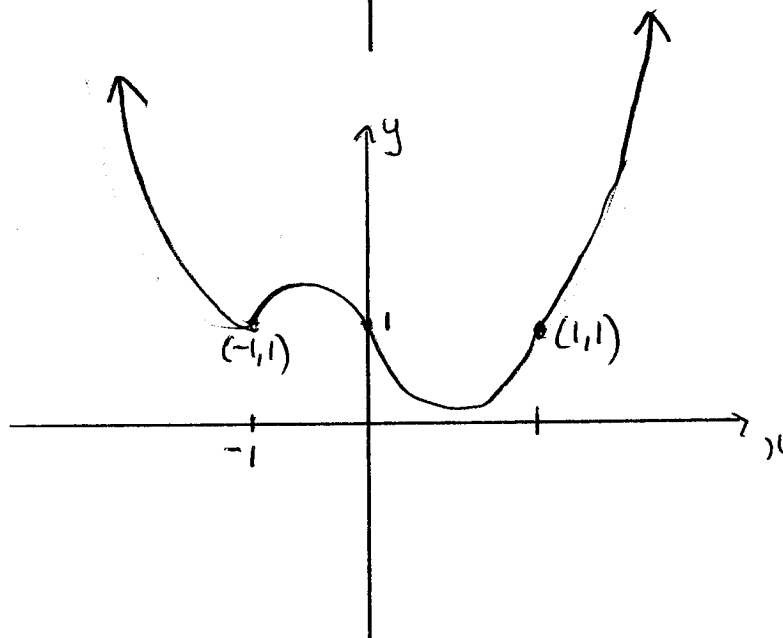
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Question 16

$$(iii) \begin{aligned} y^2 &= f(x) \\ y &= \pm \sqrt{f(x)} \end{aligned}$$



$$(iv) y = e^{f(x)}$$



$$(b) F(p) = \lim_{t \rightarrow \infty} \int_0^t x^{p-1} e^{-x} dx$$

$$(i) F(1) = \lim_{t \rightarrow \infty} \int_0^t x^0 e^{-x} dx$$

$$= \lim_{t \rightarrow \infty} [-e^{-x}]_0^t$$

$$= \lim_{t \rightarrow \infty} [-e^{-t} + 1]$$

$$= 1$$

as required.

2

2

- ① correct
integration
and sub

- ① $\lim_{t \rightarrow \infty} e^{-t} = 0$.

12

Question 16

$$(ii) F(p+1) = pF(p).$$

$$\text{LHS} = \lim_{t \rightarrow \infty} \int_0^t x^{p+1-1} e^{-x} dx$$

$$= \lim_{t \rightarrow \infty} \int_0^t x^p e^{-x} dx$$

$$= \lim_{t \rightarrow \infty} \left[-e^{-x} x^p \right]_0^t - \int_0^t p x^{p-1} e^{-x} dx$$

$$= \lim_{t \rightarrow \infty} \left[-e^{-t} t^p - 0 + p \int_0^t x^{p-1} e^{-x} dx \right]$$

$$= \lim_{t \rightarrow \infty} \left[0 + p \int_0^t x^{p-1} e^{-x} dx \right]$$

$$= p \times \lim_{t \rightarrow \infty} \int_0^t x^{p-1} e^{-x} dx$$

$$= pF(p)$$

$$= \text{RHS.}$$

- ① correct expression for $F(p+1)$ or equivalent

- ① correct use of int. parts

- ① correct execution and sub in of $F(p)$

3

iii

Question 16

$$F(n+1) = n F(n)$$

$$\therefore F(n) = (n-1) F(n-1)$$

$$= (n-1)(n-2) F(n-2)$$

$$= (n-1)(n-2)(n-3) F(n-3)$$

$$= (n-1)(n-2)(n-3) \dots 1$$

$$\boxed{F(n) = (n-1)!}$$

— ① Showing
any pattern
from
 $F(n+1) = n F(n)$

— ① Answer

2